

Space_Missions_Analysis_(start)

January 28, 2026

1 Introduction

This dataset was scraped from nextspaceflight.com and includes all the space missions since the beginning of Space Race between the USA and the Soviet Union in 1957!

1.0.1 Install Package with Country Codes

```
[75]: # pip install iso3166
```

1.0.2 Upgrade Plotly

Run the cell below if you are working with Google Colab.

```
[76]: # %pip install --upgrade plotly
```

1.0.3 Import Statements

```
[77]: import numpy as np
import pandas as pd
import plotly.express as px
import matplotlib.pyplot as plt
import seaborn as sns

# These might be helpful:
from iso3166 import countries
from datetime import datetime, timedelta
```

1.0.4 Notebook Presentation

```
[78]: pd.options.display.float_format = '{:,.2f}'.format
```

1.0.5 Load the Data

```
[79]: df_data = pd.read_csv('mission_launches.csv')
```

2 Preliminary Data Exploration

- What is the shape of df_data?

- How many rows and columns does it have?
- What are the column names?
- Are there any NaN values or duplicates?

```
[80]: df_data.shape
```

```
[80]: (4324, 9)
```

There are 4324 rows and 9 columns

```
[81]: print(f'The columns names are : {df_data.columns.values}')
```

```
The columns names are : ['Unnamed: 0.1' 'Unnamed: 0' 'Organisation' 'Location'
'Date' 'Detail'
'Rocket_Status' 'Price' 'Mission_Status']
```

```
[82]: print(f'There are NaN values : {df_data.isna().values.any()}')
```

There are NaN values : True

```
[83]: df_data.isna().any()
```

```
[83]: Unnamed: 0.1      False
      Unnamed: 0      False
      Organisation    False
      Location        False
      Date            False
      Detail          False
      Rocket_Status   False
      Price           True
      Mission_Status  False
      dtype: bool
```

```
[84]: print(f'There are duplicated values : {df_data.duplicated().values.any()}')
```

There are duplicated values : False

2.1 Data Cleaning - Check for Missing Values and Duplicates

Consider removing columns containing junk data.

```
[85]: df_data.head()
```

```
[85]:   Unnamed: 0.1  Unnamed: 0  Organisation \
0             0           0         SpaceX
1             1           1           CASC
2             2           2         SpaceX
3             3           3        Roscosmos
4             4           4           ULA
```

	Location \
0	LC-39A, Kennedy Space Center, Florida, USA
1	Site 9401 (SLS-2), Jiuquan Satellite Launch Ce...
2	Pad A, Boca Chica, Texas, USA
3	Site 200/39, Baikonur Cosmodrome, Kazakhstan
4	SLC-41, Cape Canaveral AFS, Florida, USA

	Date	Detail \
0	Fri Aug 07, 2020 05:12 UTC	Falcon 9 Block 5 Starlink V1 L9 & BlackSky
1	Thu Aug 06, 2020 04:01 UTC	Long March 2D Gaofen-9 04 & Q-SAT
2	Tue Aug 04, 2020 23:57 UTC	Starship Prototype 150 Meter Hop
3	Thu Jul 30, 2020 21:25 UTC	Proton-M/Briz-M Ekspress-80 & Ekspress-103
4	Thu Jul 30, 2020 11:50 UTC	Atlas V 541 Perseverance

	Rocket_Status	Price	Mission_Status
0	StatusActive	50.0	Success
1	StatusActive	29.75	Success
2	StatusActive	NaN	Success
3	StatusActive	65.0	Success
4	StatusActive	145.0	Success

```
[86]: df_data.tail()
```

```
[86]:
```

	Unnamed: 0.1	Unnamed: 0	Organisation \
4319	4319	4319	US Navy
4320	4320	4320	AMBA
4321	4321	4321	US Navy
4322	4322	4322	RVSN USSR
4323	4323	4323	RVSN USSR

	Location	Date \
4319	LC-18A, Cape Canaveral AFS, Florida, USA	Wed Feb 05, 1958 07:33 UTC
4320	LC-26A, Cape Canaveral AFS, Florida, USA	Sat Feb 01, 1958 03:48 UTC
4321	LC-18A, Cape Canaveral AFS, Florida, USA	Fri Dec 06, 1957 16:44 UTC
4322	Site 1/5, Baikonur Cosmodrome, Kazakhstan	Sun Nov 03, 1957 02:30 UTC
4323	Site 1/5, Baikonur Cosmodrome, Kazakhstan	Fri Oct 04, 1957 19:28 UTC

	Detail	Rocket_Status	Price	Mission_Status
4319	Vanguard Vanguard TV3BU	StatusRetired	NaN	Failure
4320	Juno I Explorer 1	StatusRetired	NaN	Success
4321	Vanguard Vanguard TV3	StatusRetired	NaN	Failure
4322	Sputnik 8K71PS Sputnik-2	StatusRetired	NaN	Success
4323	Sputnik 8K71PS Sputnik-1	StatusRetired	NaN	Success

```
[87]: df_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4324 entries, 0 to 4323
```

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Unnamed: 0.1	4324 non-null	int64
1	Unnamed: 0	4324 non-null	int64
2	Organisation	4324 non-null	object
3	Location	4324 non-null	object
4	Date	4324 non-null	object
5	Detail	4324 non-null	object
6	Rocket_Status	4324 non-null	object
7	Price	964 non-null	object
8	Mission_Status	4324 non-null	object

dtypes: int64(2), object(7)

memory usage: 304.2+ KB

```
[88]: df_data.drop(['Unnamed: 0', 'Unnamed: 0.1'], inplace=True, axis = 1)
df_data.head()
```

```
[88]: Organisation                                Location \
0      SpaceX                LC-39A, Kennedy Space Center, Florida, USA
1      CASC   Site 9401 (SLS-2), Jiuquan Satellite Launch Ce...
2      SpaceX                                Pad A, Boca Chica, Texas, USA
3      Roscosmos        Site 200/39, Baikonur Cosmodrome, Kazakhstan
4      ULA                SLC-41, Cape Canaveral AFS, Florida, USA

                                Date                                Detail \
0  Fri Aug 07, 2020 05:12 UTC  Falcon 9 Block 5 | Starlink V1 L9 & BlackSky
1  Thu Aug 06, 2020 04:01 UTC                                Long March 2D | Gaofen-9 04 & Q-SAT
2  Tue Aug 04, 2020 23:57 UTC                                Starship Prototype | 150 Meter Hop
3  Thu Jul 30, 2020 21:25 UTC  Proton-M/Briz-M | Ekspress-80 & Ekspress-103
4  Thu Jul 30, 2020 11:50 UTC                                Atlas V 541 | Perseverance

Rocket_Status  Price  Mission_Status
0  StatusActive   50.0          Success
1  StatusActive  29.75          Success
2  StatusActive   NaN          Success
3  StatusActive   65.0          Success
4  StatusActive  145.0          Success
```

```
[89]: df_data["Date"] = pd.to_datetime(
    df_data["Date"],
    format="mixed",
    errors="coerce",
    utc=True
)
```

```
[90]: print("Unparsed dates:", df_data["Date"].isna().sum())
df_data.loc[df_data["Date"].isna(), ["Date"]].head(10)
```

Unparsed dates: 0

```
[90]: Empty DataFrame
      Columns: [Date]
      Index: []
```

```
[91]: df_data.Date
```

```
[91]: 0      2020-08-07 05:12:00+00:00
      1      2020-08-06 04:01:00+00:00
      2      2020-08-04 23:57:00+00:00
      3      2020-07-30 21:25:00+00:00
      4      2020-07-30 11:50:00+00:00
      ...
      4319   1958-02-05 07:33:00+00:00
      4320   1958-02-01 03:48:00+00:00
      4321   1957-12-06 16:44:00+00:00
      4322   1957-11-03 02:30:00+00:00
      4323   1957-10-04 19:28:00+00:00
      Name: Date, Length: 4324, dtype: datetime64[ns, UTC]
```

```
[92]: df_data.Price = df_data.Price.apply(lambda x: str(x).replace(',', '.')).
      ↪astype('float64')
```

```
[93]: df_data.Price
```

```
[93]: 0      50.00
      1      29.75
      2      NaN
      3      65.00
      4     145.00
      ...
      4319     NaN
      4320     NaN
      4321     NaN
      4322     NaN
      4323     NaN
      Name: Price, Length: 4324, dtype: float64
```

```
[94]: df_data.Price.isna().sum()
```

```
[94]: 3360
```

2.2 Descriptive Statistics

```
[95]: df_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4324 entries, 0 to 4323
```

```
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Organisation           4324 non-null  object
1   Location                4324 non-null  object
2   Date                   4324 non-null  datetime64[ns, UTC]
3   Detail                 4324 non-null  object
4   Rocket_Status          4324 non-null  object
5   Price                  964 non-null   float64
6   Mission_Status         4324 non-null  object
dtypes: datetime64[ns, UTC](1), float64(1), object(5)
memory usage: 236.6+ KB
```

```
[96]: df_data.describe()
```

```
[96]:          Price
count    964.00
mean     153.79
std      288.45
min        5.30
25%       40.00
50%       62.00
75%      164.00
max     5,000.00
```

3 Number of Launches per Company

Create a chart that shows the number of space mission launches by organisation.

```
[97]: launches_per_company = df_data.Organisation.value_counts()
launches_per_company
```

```
[97]: Organisation
RVSN USSR           1777
Arianespace         279
CASC                 251
General Dynamics    251
NASA                 203
VKS RF              201
US Air Force        161
ULA                  140
Boeing              136
Martin Marietta     114
SpaceX              100
MHI                  84
Northrop             83
Lockheed             79
```

ISRO	76
Roscosmos	55
ILS	46
Sea Launch	36
ISAS	30
Kosmotras	22
US Navy	17
ISA	13
Rocket Lab	13
Eurockot	13
ESA	13
Blue Origin	12
IAI	11
ExPace	10
ASI	9
CNES	8
AMBA	8
MITT	7
JAXA	7
Land Launch	7
UT	5
KCST	5
CASIC	5
Exos	4
CECLES	4
Arm??e de l'Air	4
KARI	3
SRC	3
AEB	3
RAE	2
OKB-586	2
Yuzhmash	2
Landspace	1
Douglas	1
EER	1
Starsem	1
Virgin Orbit	1
IRGC	1
i-Space	1
OneSpace	1
Sandia	1
Khrunichev	1

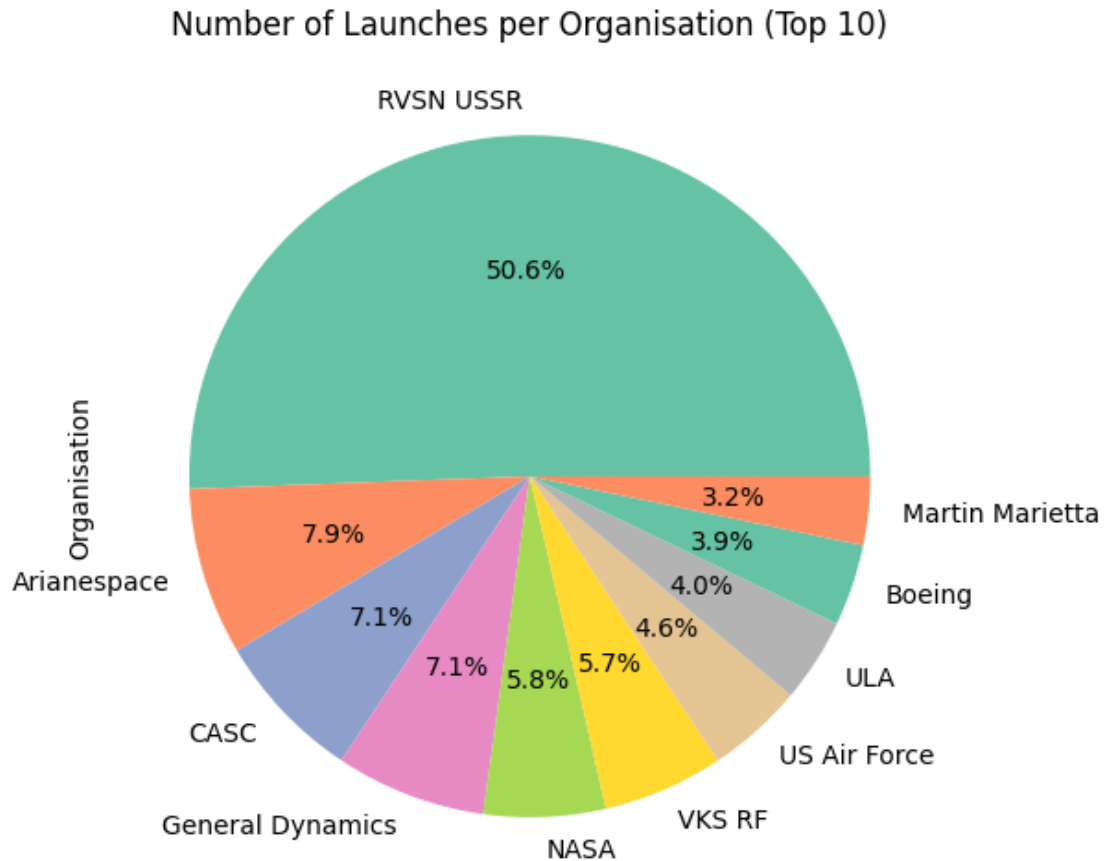
Name: count, dtype: int64

```
[98]: color_palette = sns.color_palette("Set2")
      launches_per_company[:10].plot(
          kind='pie',
```

```

title='Number of Launches per Organisation (Top 10)',
figsize=(6,6),
autopct='%1.1f%%',
colors=color_palette
)
plt.ylabel('Organisation') # Set the y-axis label
plt.show()

```



4 Number of Active versus Retired Rockets

How many rockets are active compared to those that are decommissioned?

```
[99]: df_data.Rocket_Status.value_counts()
```

```

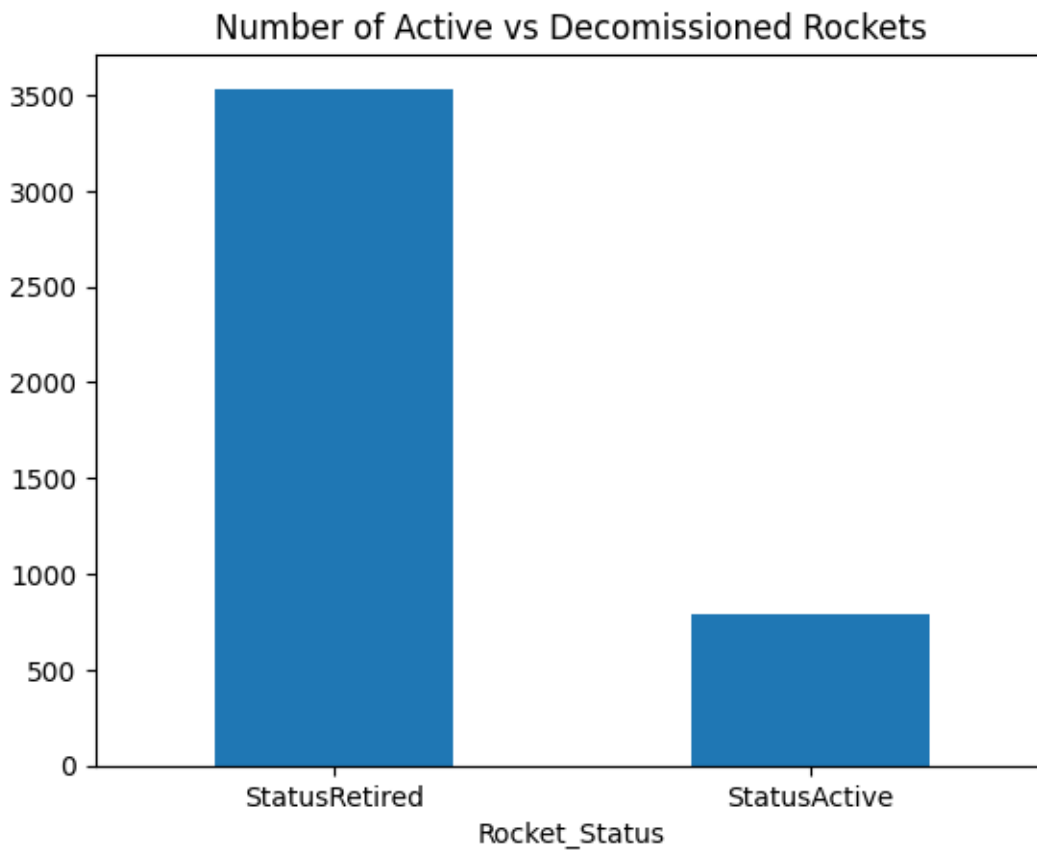
[99]: Rocket_Status
      StatusRetired    3534
      StatusActive    790

```


Name: count, dtype: int64

```
[100]: df_data.Rocket_Status.value_counts().plot(
        kind='bar',
        title='Number of Active vs Decomissioned Rockets',
        rot=0,
    )
```

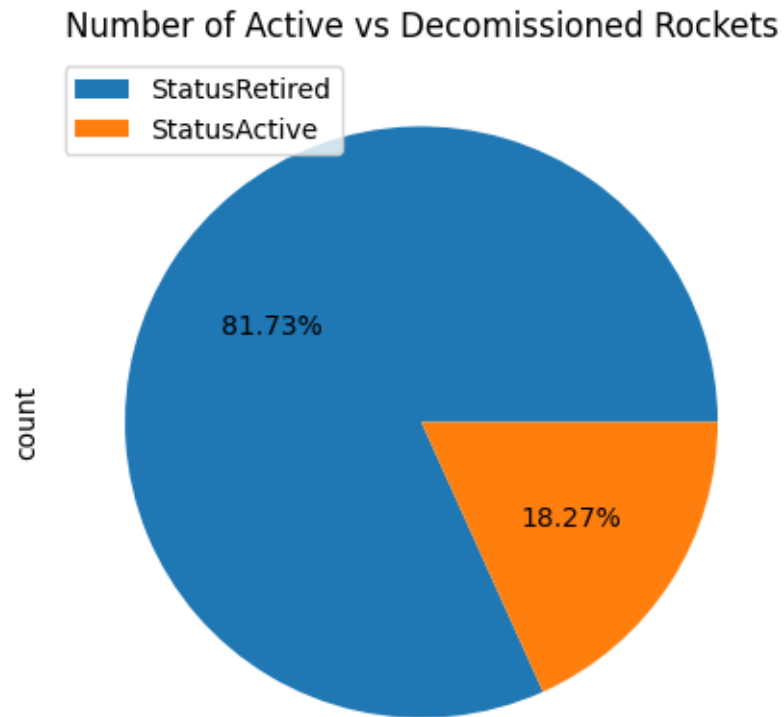
```
[100]: <Axes: title={'center': 'Number of Active vs Decomissioned Rockets'},
      xlabel='Rocket_Status'>
```



```
[101]: df_data.Rocket_Status.value_counts().plot(
        kind='pie',
        title='Number of Active vs Decomissioned Rockets',
        figsize = (5,5),
        autopct = '%1.2f%%',
        labels = None
    )

plt.legend(labels = df_data.Rocket_Status.value_counts().index)
```

```
plt.show()
```



18.27% of the rockets are still active.

5 Distribution of Mission Status

How many missions were successful? How many missions failed?

```
[102]: mission_status_df = df_data.Mission_Status.value_counts()
```

```
[103]: mission_status_df
```

```
[103]: Mission_Status
Success          3879
Failure           339
Partial Failure   102
Prelaunch Failure  4
Name: count, dtype: int64
```

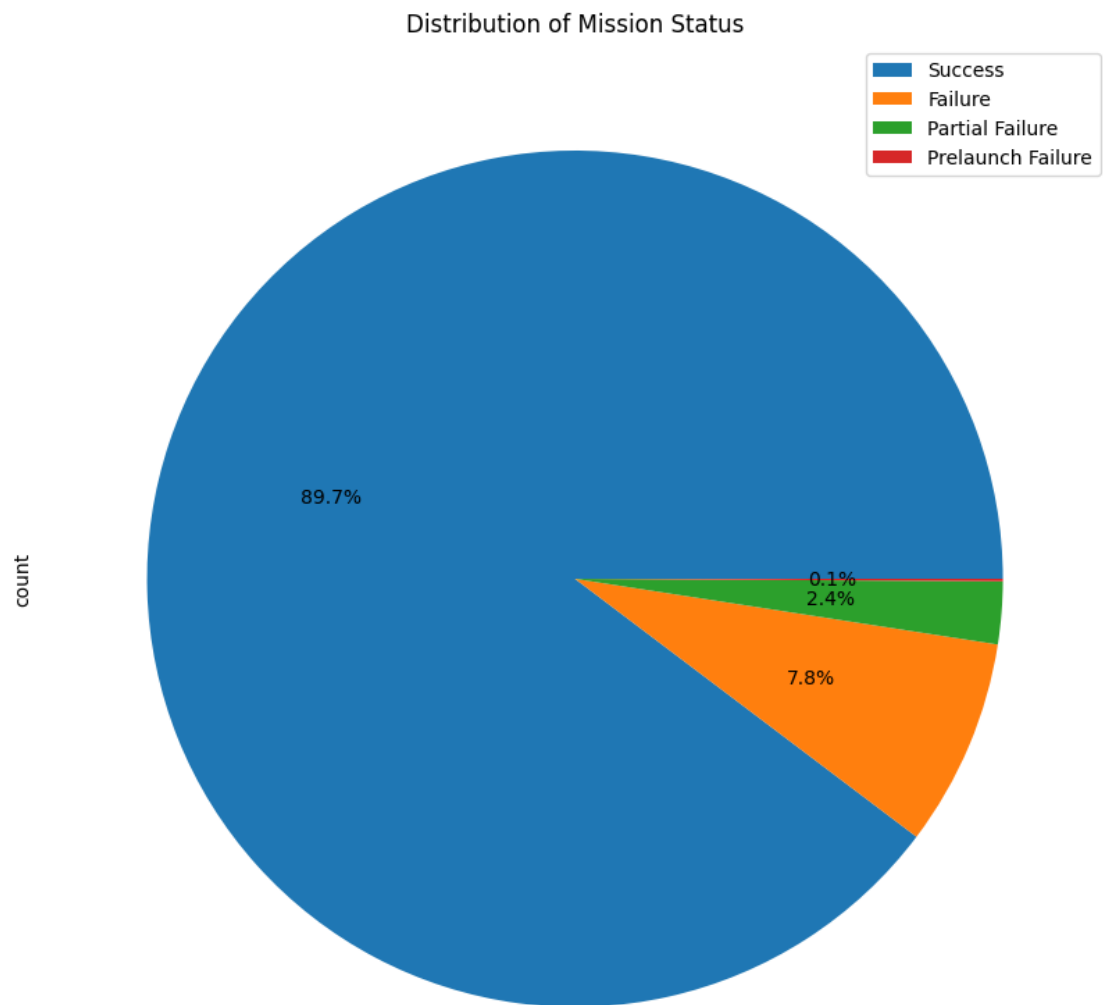
```
[104]: mission_status_df.plot(
        kind='pie',
```

```

figsize=(10,10),
autopct='%1.1f%%',
title = 'Distribution of Mission Status',
labels=None
)

plt.legend(labels=mission_status_df.index)
plt.show()

```



89.7% of the missions were succesful while 7.8% failed to launch.

6 How Expensive are the Launches?

Create a histogram and visualise the distribution. The price column is given in USD millions (careful of missing values).

```
[105]: cost_distribution = df_data.Price.value_counts()
```

```
[106]: cost_distribution
```

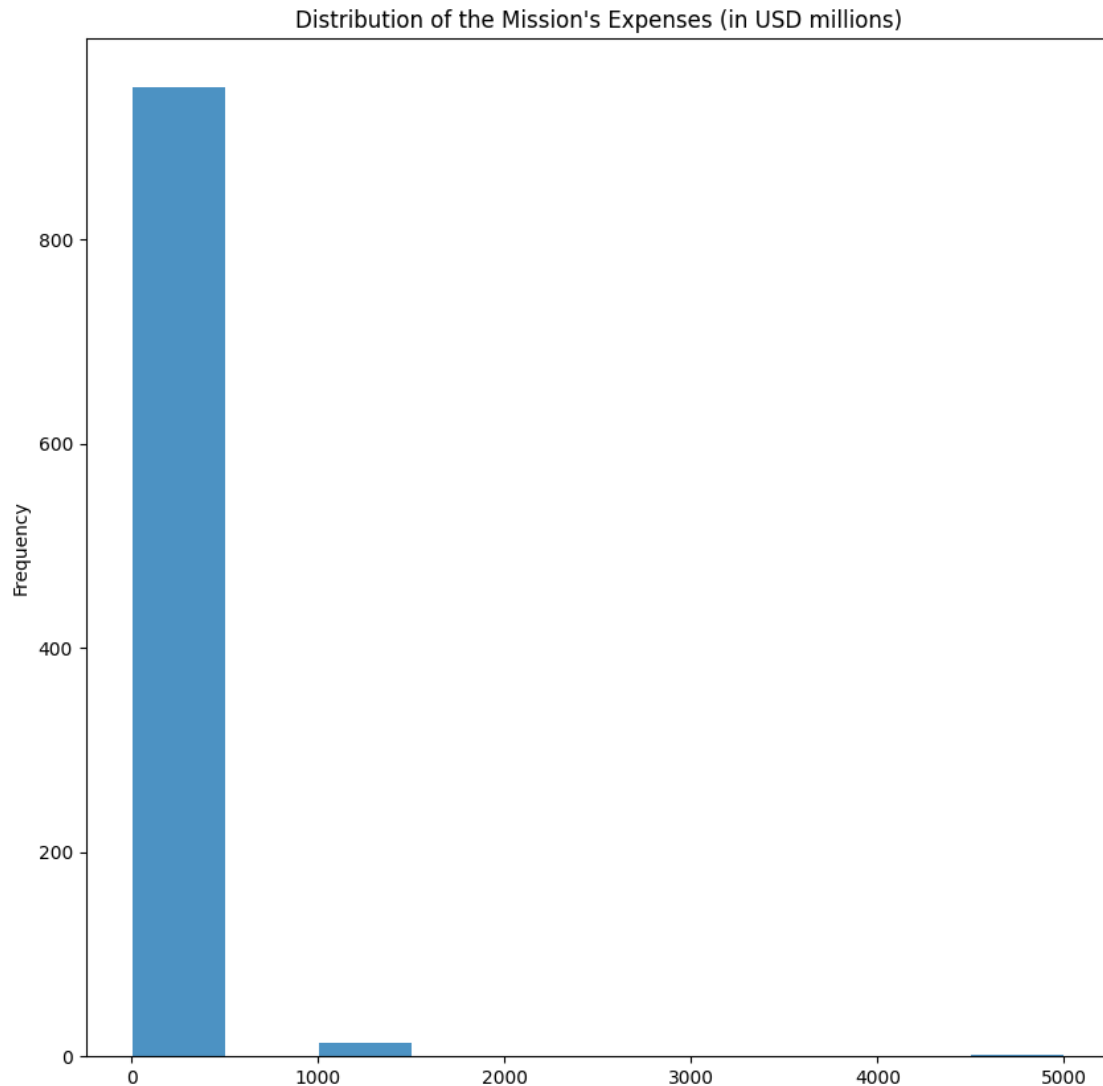
```
[106]: Price
450.00      136
200.00       75
40.00       55
62.00       41
30.80       38
109.00      37
50.00       34
64.68       34
29.75       33
90.00       32
41.80       31
48.50       26
29.15       25
31.00       22
29.00       22
59.00       22
69.70       17
21.00       16
65.00       16
35.00       16
56.50       15
37.00       15
164.00       15
7.50        14
1,160.00    13
47.00       13
25.00       12
350.00      11
153.00      11
45.00       10
112.50       9
5.30         9
123.00       8
145.00       7
85.00        7
120.00       7
80.00        7
115.00       6
```

59.50	5
7.00	5
46.00	5
136.60	4
63.23	4
140.00	3
133.00	3
190.00	3
130.00	3
135.00	2
5,000.00	2
39.00	2
55.00	1
15.00	1
20.14	1
20.00	1
12.00	1
28.30	1

Name: count, dtype: int64

```
[107]: df_data.Price.plot(
        kind='hist',
        title = "Distribution of the Mission's Expenses (in USD millions)",
        figsize = (10,10),
        alpha=0.8
    )
```

```
[107]: <Axes: title={'center': "Distribution of the Mission's Expenses (in USD
millions)"}, ylabel='Frequency'>
```



[]:

7 Use a Choropleth Map to Show the Number of Launches by Country

- Create a choropleth map using [the plotly documentation](#)
- Experiment with [plotly's available colours](#). I quite like the sequential colour **matter** on this map.
- You'll need to extract a **country** feature as well as change the country names that no longer exist.

Wrangle the Country Names

You'll need to use a 3 letter country code for each country. You might have to change some country

names.

- Russia is the Russian Federation
- New Mexico should be USA
- Yellow Sea refers to China
- Shahrud Missile Test Site should be Iran
- Pacific Missile Range Facility should be USA
- Barents Sea should be Russian Federation
- Gran Canaria should be USA

You can use the iso3166 package to convert the country names to Alpha3 format.

```
[108]: df_data['Country'] = df_data['Location'].apply(lambda x: x.split(',')[1].  
↳strip())  
df_data['Country'].value_counts()
```

```
[108]: Country  
Russia                1395  
USA                   1344  
Kazakhstan            701  
France                303  
China                 268  
Japan                 126  
India                 76  
Pacific Ocean         36  
New Zealand           13  
Iran                  13  
Israel                11  
Kenya                  9  
Australia              6  
North Korea            5  
New Mexico             4  
South Korea            3  
Barents Sea            3  
Brazil                 3  
Gran Canaria           2  
Pacific Missile Range Facility 1  
Yellow Sea             1  
Shahrud Missile Test Site 1  
Name: count, dtype: int64
```

```
[109]: df_data['Country'].replace(  
    {  
        'Russia': 'Russian Federation',  
        'Iran': 'Iran, Islamic Republic of',  
        'New Mexico': 'USA',  
        'Barents Sea': 'Russian Federation',  
        'Gran Canaria': 'USA',
```

```

        'Yellow Sea': 'China',
        'Shahrud Missile Test Site': 'Iran, Islamic Republic of',
        'Pacific Missile Range Facility': 'USA',
        'North Korea': "Korea, Democratic People's Republic of",
        'Pacific Ocean': 'Kiribati',
        'South Korea': 'Korea, Republic of'
    },
    inplace=True
)
df_data['Country'] = df_data['Country'].apply(lambda x: countries.get(x).alpha3)
df_data['Country'].value_counts()

```

C:\Users\user\AppData\Local\Temp\ipykernel_22340\3590054349.py:1: FutureWarning:

A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```

[109]: Country
RUS    1398
USA    1351
KAZ     701
FRA     303
CHN     269
JPN     126
IND      76
KIR      36
IRN      14
NZL      13
ISR      11
KEN       9
AUS       6
PRK       5
KOR       3
BRA       3
Name: count, dtype: int64

```

```

[110]: launches_per_country = df_data['Country'].value_counts()
launches_per_country

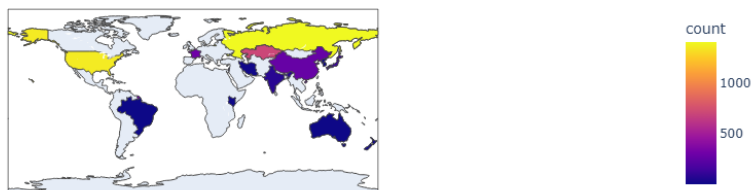
```



```
[110]: Country
      RUS    1398
      USA    1351
      KAZ     701
      FRA     303
      CHN     269
      JPN     126
      IND      76
      KIR      36
      IRN      14
      NZL      13
      ISR      11
      KEN       9
      AUS       6
      PRK       5
      KOR       3
      BRA       3
      Name: count, dtype: int64
```

```
[111]: px.choropleth(
        launches_per_country,
        locations=launches_per_country.index,
        color=launches_per_country,
        title = 'Number of Launches by Country'
    )
```

Number of Launches by Country



8 Use a Choropleth Map to Show the Number of Failures by Country

```
[112]: failures_per_country = df_data[df_data['Mission_Status'] == 'Failure']
        failures_per_country['Country'].value_counts()
        failures_per_country
```

```
[112]: Country
      USA    129
      KAZ     72
      RUS     63
      CHN     19
      FRA     13
      JPN     10
      IRN      8
      IND      8
      KIR      3
      PRK      3
      AUS      3
      NZL      2
      KOR      2
      ISR      2
      BRA      2
      Name: count, dtype: int64
```

```
[113]: px.choropleth(
        failures_per_country,
        locations=failures_per_country.index,
        color=failures_per_country,
        title = 'Number of Failures by Country'
    )
```

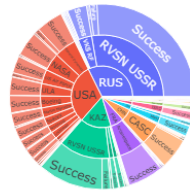
Number of Failures by Country



9 Create a Plotly Sunburst Chart of the countries, organisations, and mission status.

```
[114]: px.sunburst(df_data, path=['Country', 'Organisation', 'Mission_Status'],
                  ↪title='Mission Status by Country and Organization')
```

Mission Status by Country and Organization



10 Analyse the Total Amount of Money Spent by Organisation on Space Missions

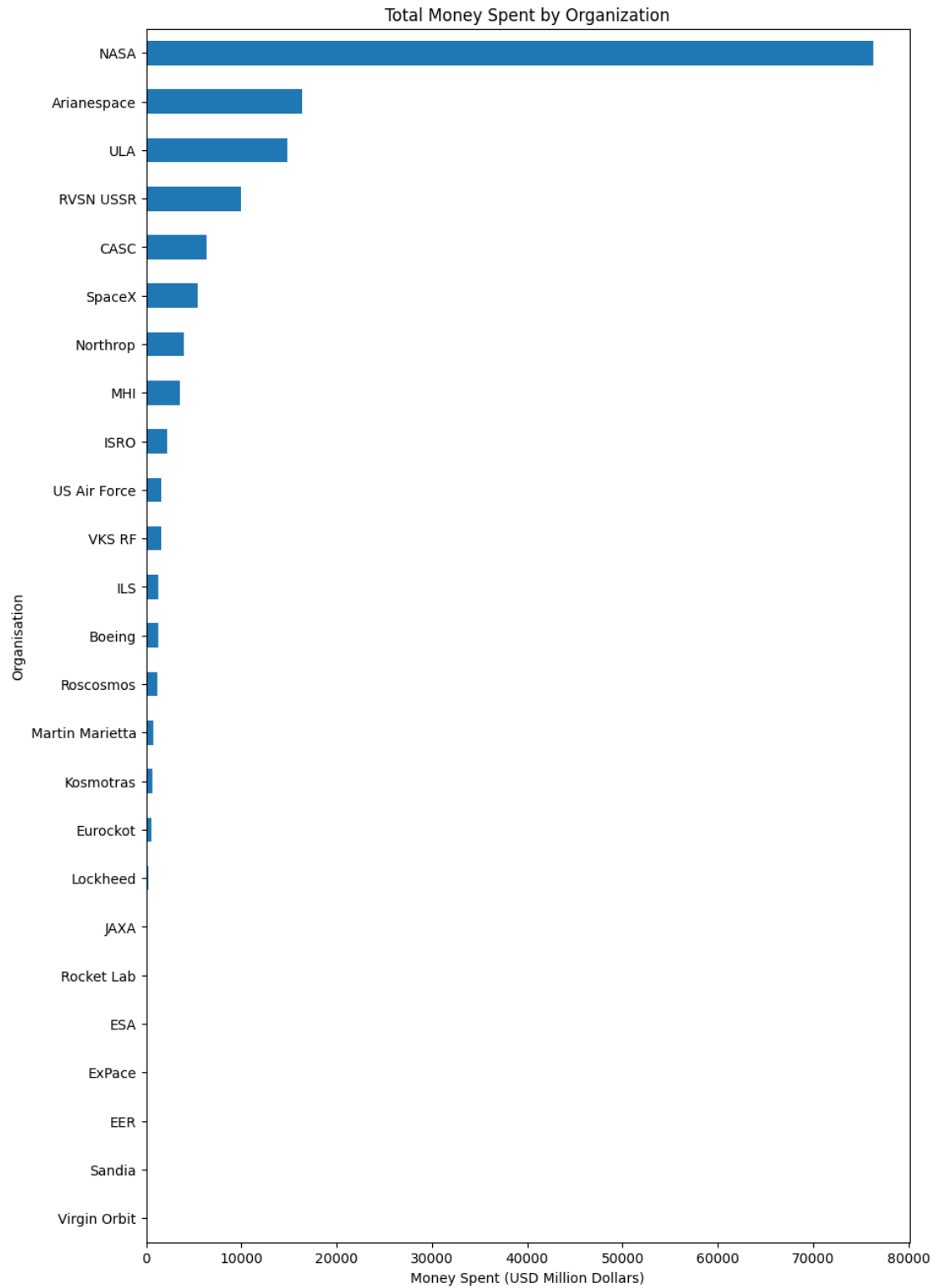
```
[115]: money_spent_by_organization = df_data.groupby('Organisation')['Price'].sum()
money_spent_by_organization[money_spent_by_organization!=0]
```

```
[115]: Organisation
Arianespace      16,345.00
Boeing            1,241.00
CASC              6,340.26
EER               20.00
ESA               37.00
Eurockot          543.40
ExPace            28.30
ILS              1,320.00
ISRO              2,177.00
JAXA              168.00
Kosmotras         638.00
Lockheed          280.00
MHI               3,532.50
Martin Marietta   721.40
NASA              76,280.00
Northrop          3,930.00
RVSN USSR         10,000.00
Rocket Lab        97.50
Roscosmos         1,187.50
Sandia            15.00
SpaceX            5,444.00
ULA              14,798.00
US Air Force      1,550.92
VKS RF            1,548.90
Virgin Orbit      12.00
```

Name: Price, dtype: float64

```
[116]: money_spent_by_organization[money_spent_by_organization!=0].sort_values().plot(
        kind='barh',
        title = 'Total Money Spent by Organization',
        figsize=(10,16)
    )

plt.xlabel('Money Spent (USD Million Dollars)')
plt.show()
```



11 Analyse the Amount of Money Spent by Organisation per Launch

```
[117]: avg_money_spent = df_data.groupby('Organisation').agg({'Price':np.mean})
avg_money_spent.dropna()
```

C:\Users\user\AppData\Local\Temp\ipykernel_22340\2928463774.py:1: FutureWarning:

The provided callable <function mean at 0x0000024366158680> is currently using SeriesGroupBy.mean. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "mean" instead.

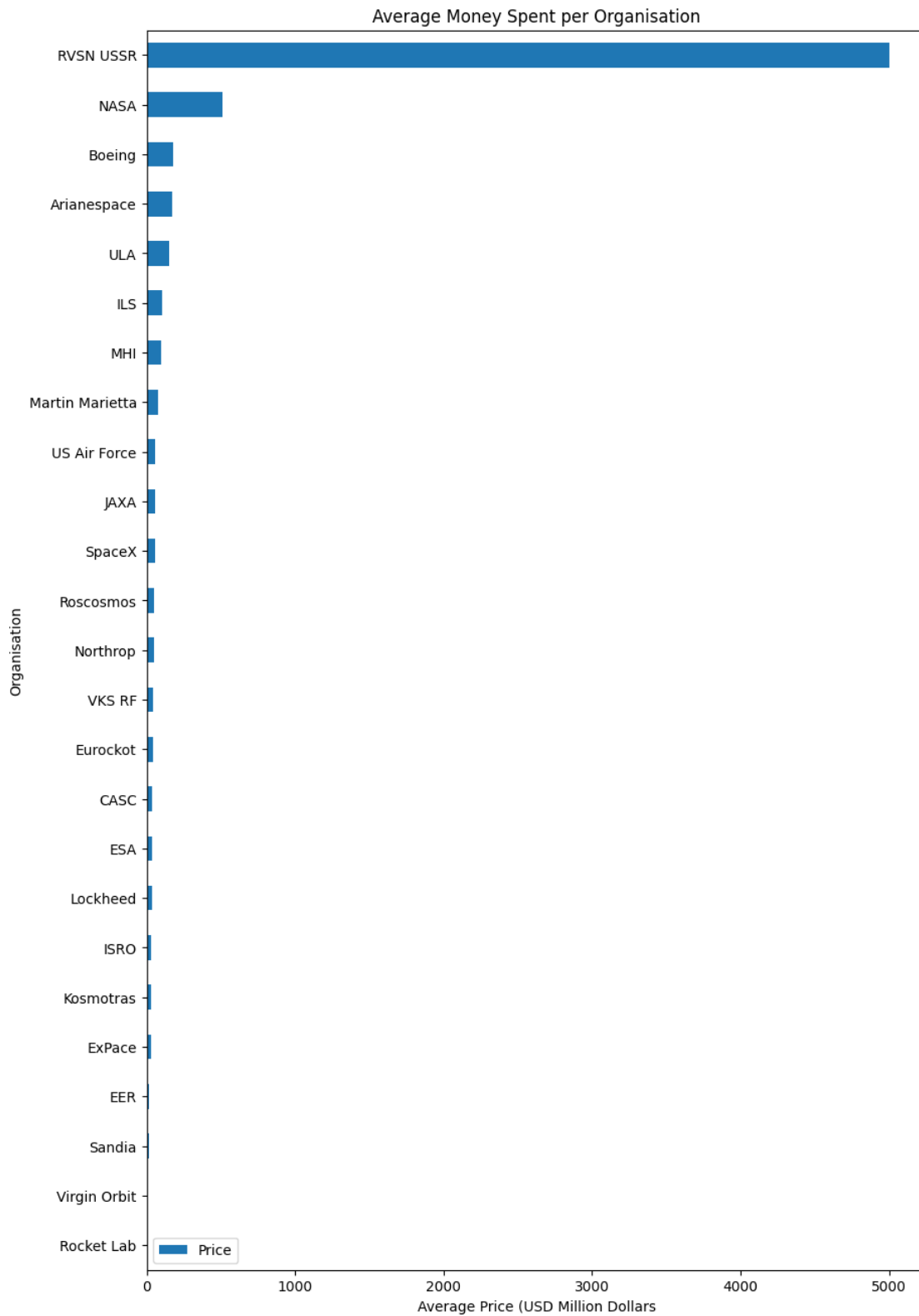
```
[117]:
```

Organisation	Price
Arianespace	170.26
Boeing	177.29
CASC	40.13
EER	20.00
ESA	37.00
Eurockot	41.80
ExPace	28.30
ILS	101.54
ISRO	32.49
JAXA	56.00
Kosmotras	29.00
Lockheed	35.00
MHI	95.47
Martin Marietta	80.16
NASA	511.95
Northrop	47.35
RVSN USSR	5,000.00
Rocket Lab	7.50
Roscosmos	51.63
Sandia	15.00
SpaceX	54.99
ULA	151.00
US Air Force	59.65
VKS RF	46.94
Virgin Orbit	12.00

```
[118]: avg_money_spent.dropna().sort_values('Price').plot(
    kind='barh',
    figsize=(10,16),
    title='Average Money Spent per Organisation',
)

plt.xlabel('Average Price (USD Million Dollars)')
```

```
plt.show()
```



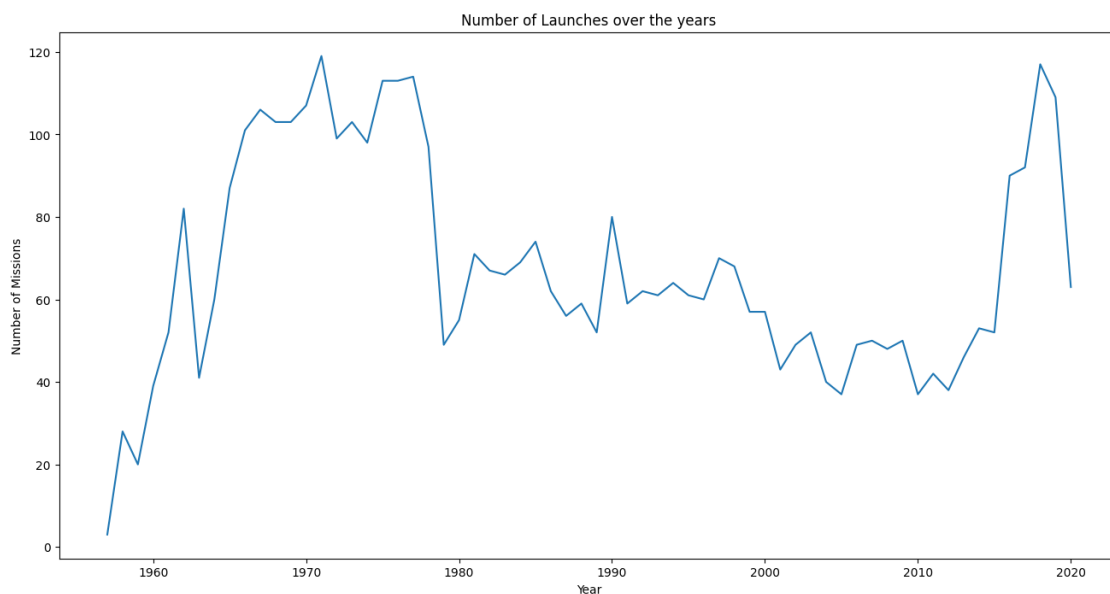
12 Chart the Number of Launches per Year

```
[119]: df_data['Year'] = df_data['Date'].dt.year
launches_per_year = df_data.groupby('Year')['Detail'].count()
launches_per_year
```

```
[119]: Year
1957      3
1958     28
1959     20
1960     39
1961     52
...
2016     90
2017     92
2018    117
2019    109
2020     63
Name: Detail, Length: 64, dtype: int64
```

```
[120]: launches_per_year.plot(
        title='Number of Launches over the years',
        figsize = (16,8)
    )

plt.ylabel('Number of Missions')
plt.show()
```



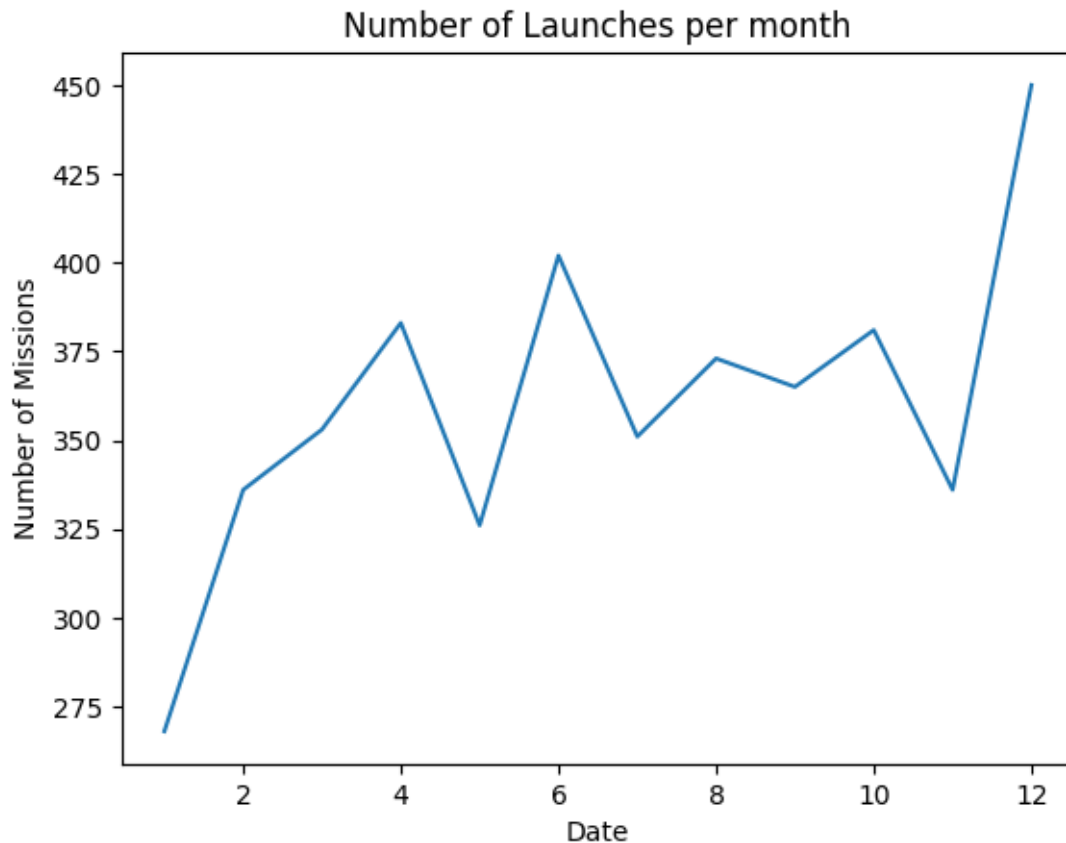
13 Chart the Number of Launches Month-on-Month until the Present

Which month has seen the highest number of launches in all time? Superimpose a rolling average on the month on month time series chart.

```
[121]: launches_per_month = df_data.groupby(df_data['Date'].dt.month)['Detail'].count()  
       launches_per_month
```

```
[121]: Date  
1      268  
2      336  
3      353  
4      383  
5      326  
6      402  
7      351  
8      373  
9      365  
10     381  
11     336  
12     450  
Name: Detail, dtype: int64
```

```
[122]: plt.figure(figsize = (16,8))  
       launches_per_month.plot(title = 'Number of Launches per month')  
  
       plt.ylabel('Number of Missions')  
       plt.show()
```



14 Launches per Month: Which months are most popular and least popular for launches?

Some months have better weather than others. Which time of year seems to be best for space missions?

```
[123]: launches_per_month[launches_per_month == launches_per_month.max()]
```

```
[123]: Date
      12    450
      Name: Detail, dtype: int64
```

```
[124]: launches_per_month[launches_per_month == launches_per_month.min()]
```

```
[124]: Date
      1    268
      Name: Detail, dtype: int64
```

It appears that most launches take place in December

15 How has the Launch Price varied Over Time?

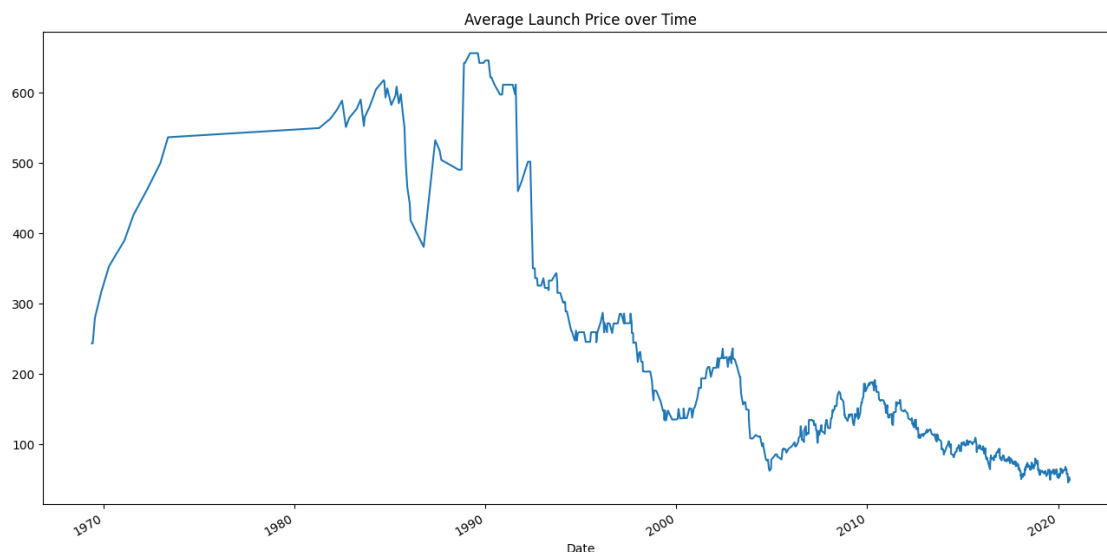
Create a line chart that shows the average price of rocket launches over time.

```
[125]: price_over_time = df_data.dropna().groupby('Date')['Price'].mean()  
price_over_time
```

```
[125]: Date  
1964-09-01 15:00:00+00:00    63.23  
1964-12-10 16:52:00+00:00    63.23  
1965-02-11 15:19:00+00:00    63.23  
1965-05-06 15:00:00+00:00    63.23  
1966-07-29 18:43:00+00:00    59.00  
  
...  
2020-07-25 03:13:00+00:00    64.68  
2020-07-30 11:50:00+00:00   145.00  
2020-07-30 21:25:00+00:00    65.00  
2020-08-06 04:01:00+00:00    29.75  
2020-08-07 05:12:00+00:00    50.00  
Name: Price, Length: 963, dtype: float64
```

```
[126]: price_over_time.rolling(30).mean().plot(  
        figsize = (16,8),  
        title = 'Average Launch Price over Time'  
    )
```

```
[126]: <Axes: title={'center': 'Average Launch Price over Time'}, xlabel='Date'>
```



16 Chart the Number of Launches over Time by the Top 10 Organisations.

How has the dominance of launches changed over time between the different players?

```
[127]: top_10_organisations = df_data['Organisation'].value_counts().
        ↪sort_values(ascending=False)[:10].index.tolist()
top_10_organisations
```

```
[127]: ['RVSN USSR',
        'Arianespace',
        'CASC',
        'General Dynamics',
        'NASA',
        'VKS RF',
        'US Air Force',
        'ULA',
        'Boeing',
        'Martin Marietta']
```

```
[128]: launches_by_top_10 = df_data[df_data['Organisation']
        ↪isin(top_10_organisations)].groupby(['Year', 'Organisation'])['Detail'].
        ↪count().unstack(level=1)
launches_by_top_10.fillna(0, inplace=True)
launches_by_top_10
```

```
[128]: Organisation  Arianespace  Boeing  CASC  General Dynamics  Martin Marietta  \
Year
1957                0.00    0.00  0.00                0.00                0.00
1958                0.00    0.00  0.00                0.00                0.00
1959                0.00    0.00  0.00                1.00                0.00
1960                0.00    0.00  0.00                5.00                0.00
1961                0.00    0.00  0.00                8.00                0.00
...
2016                11.00    0.00 22.00                0.00                0.00
2017                11.00    0.00 16.00                0.00                0.00
2018                11.00    0.00 37.00                0.00                0.00
2019                 9.00    0.00 27.00                0.00                0.00
2020                 4.00    0.00 19.00                0.00                0.00

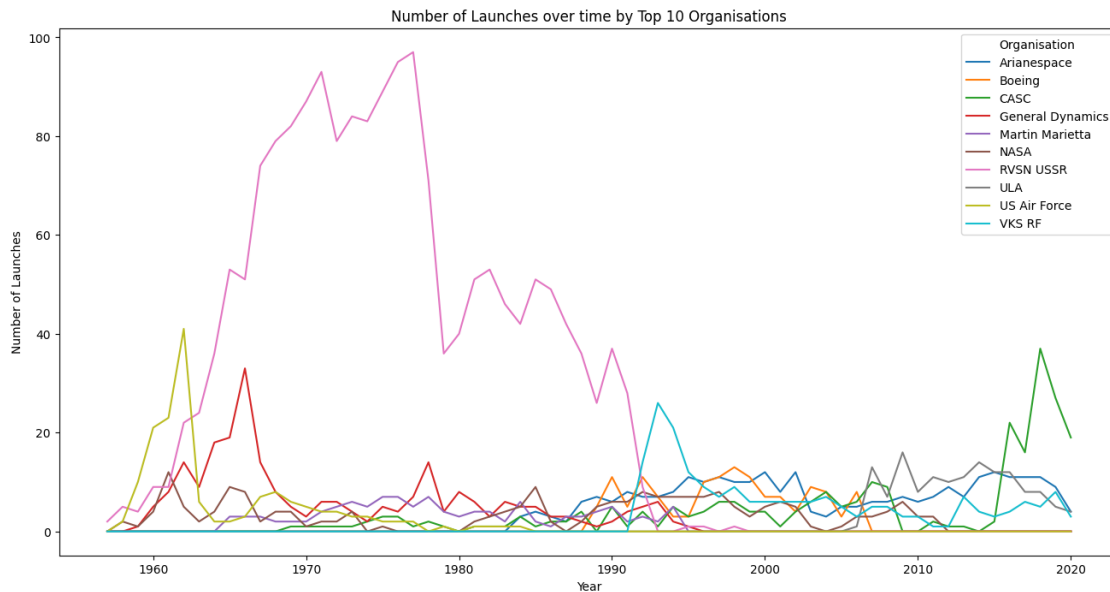
Organisation  NASA  RVSN USSR   ULA  US Air Force  VKS RF
Year
1957          0.00    2.00  0.00          0.00    0.00
1958          2.00    5.00  0.00          2.00    0.00
1959          1.00    4.00  0.00         10.00    0.00
1960          4.00    9.00  0.00         21.00    0.00
1961         12.00    9.00  0.00         23.00    0.00
```

...	...	...	...	...	...
2016	0.00	0.00	12.00	0.00	4.00
2017	0.00	0.00	8.00	0.00	6.00
2018	0.00	0.00	8.00	0.00	5.00
2019	0.00	0.00	5.00	0.00	8.00
2020	0.00	0.00	4.00	0.00	3.00

[64 rows x 10 columns]

```
[129]: launches_by_top_10.plot(
        figsize=(16,8),
        title = 'Number of Launches over time by Top 10 Organisations',
    )

plt.ylabel('Number of Launches')
plt.show()
```



USAF was realising the most launches up until the beginning of the 60s when RVSN USSR took helm on the race until the beginning of the 90s. That's when the RVSN rapidly lost ground up until they completely stopped launching. After that Arianespace, Boeing and VKS RF seemed to lead the race, up until mid 2010s when CASC surpassed them and has been leading since.

17 Cold War Space Race: USA vs USSR

The cold war lasted from the start of the dataset up until 1991.

```
[130]: cold_war_df = df_data[df_data['Year']<=1991].sort_values('Year').
        ↪reset_index(drop=True)
```

```
[131]: cold_war_df.head()
```

```
[131]:
```

	Organisation	Location \
0	RVSN USSR	Site 1/5, Baikonur Cosmodrome, Kazakhstan
1	RVSN USSR	Site 1/5, Baikonur Cosmodrome, Kazakhstan
2	US Navy	LC-18A, Cape Canaveral AFS, Florida, USA
3	AMBA	LC-26A, Cape Canaveral AFS, Florida, USA
4	US Air Force	LC-11, Cape Canaveral AFS, Florida, USA

	Date	Detail	Rocket_Status	Price \
0	1957-10-04 19:28:00+00:00	Sputnik 8K71PS Sputnik-1	StatusRetired	NaN
1	1957-11-03 02:30:00+00:00	Sputnik 8K71PS Sputnik-2	StatusRetired	NaN
2	1957-12-06 16:44:00+00:00	Vanguard Vanguard TV3	StatusRetired	NaN
3	1958-02-01 03:48:00+00:00	Juno I Explorer 1	StatusRetired	NaN
4	1958-12-18 23:02:00+00:00	SM-65B Atlas SCORE	StatusRetired	NaN

	Mission_Status	Country	Year
0	Success	KAZ	1957
1	Success	KAZ	1957
2	Failure	USA	1957
3	Success	USA	1958
4	Success	USA	1958

```
[132]: cold_war_df.shape
```

```
[132]: (2607, 9)
```

```
[133]: cold_war_df['Country'].duplicated().any()
```

```
[133]: True
```

```
[134]: for country in cold_war_df['Country'].drop_duplicates():
        print(countries.get(country))
```

```
Country(name='Kazakhstan', alpha2='KZ', alpha3='KAZ', numeric='398',
apolitical_name='Kazakhstan')
Country(name='United States of America', alpha2='US', alpha3='USA',
numeric='840', apolitical_name='United States of America')
Country(name='Russian Federation', alpha2='RU', alpha3='RUS', numeric='643',
apolitical_name='Russian Federation')
Country(name='France', alpha2='FR', alpha3='FRA', numeric='250',
apolitical_name='France')
Country(name='Japan', alpha2='JP', alpha3='JPN', numeric='392',
apolitical_name='Japan')
Country(name='Australia', alpha2='AU', alpha3='AUS', numeric='036',
```

```

apolitical_name='Australia')
Country(name='Kenya', alpha2='KE', alpha3='KEN', numeric='404',
apolitical_name='Kenya')
Country(name='China', alpha2='CN', alpha3='CHN', numeric='156',
apolitical_name='China')
Country(name='India', alpha2='IN', alpha3='IND', numeric='356',
apolitical_name='India')
Country(name='Israel', alpha2='IL', alpha3='ISR', numeric='376',
apolitical_name='Israel')

```

17.1 Create a Plotly Pie Chart comparing the total number of launches of the USSR and the USA

Hint: Remember to include former Soviet Republics like Kazakhstan when analysing the total number of launches.

```

[135]: cold_war_df['Country'].replace(
        {'KAZ': 'RUS'},
        inplace=True
    )
cold_war_df['Country'].value_counts()

```

C:\Users\user\AppData\Local\Temp\ipykernel_22340\2180976691.py:1: FutureWarning:

A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```

[135]: Country
RUS      1770
USA       662
FRA        61
JPN        52
CHN        38
KEN         9
IND         7
AUS         6
ISR         2
Name: count, dtype: int64

```

```
[136]: ussr_vs_usa = cold_war_df[cold_war_df['Country'].isin(['RUS', 'USA'])]
       ussr_vs_usa
```

```
[136]:
```

	Organisation	Location \
0	RVSN USSR	Site 1/5, Baikonur Cosmodrome, Kazakhstan
1	RVSN USSR	Site 1/5, Baikonur Cosmodrome, Kazakhstan
2	US Navy	LC-18A, Cape Canaveral AFS, Florida, USA
3	AMBA	LC-26A, Cape Canaveral AFS, Florida, USA
4	US Air Force	LC-11, Cape Canaveral AFS, Florida, USA
...	...	...
2601	RVSN USSR	Site 32/2, Plesetsk Cosmodrome, Russia
2602	NASA	LC-39B, Kennedy Space Center, Florida, USA
2603	RVSN USSR	Site 133/3, Plesetsk Cosmodrome, Russia
2604	RVSN USSR	Site 32/2, Plesetsk Cosmodrome, Russia
2605	RVSN USSR	Site 132/1, Plesetsk Cosmodrome, Russia

	Date	Detail \
0	1957-10-04 19:28:00+00:00	Sputnik 8K71PS Sputnik-1
1	1957-11-03 02:30:00+00:00	Sputnik 8K71PS Sputnik-2
2	1957-12-06 16:44:00+00:00	Vanguard Vanguard TV3
3	1958-02-01 03:48:00+00:00	Juno I Explorer 1
4	1958-12-18 23:02:00+00:00	SM-65B Atlas SCORE
...	...	...
2601	1991-06-04 09:00:00+00:00	Tsyklon-3 Okean 3
2602	1991-06-05 13:24:00+00:00	Space Shuttle Columbia STS-40
2603	1991-06-11 05:42:00+00:00	Cosmos-3M (11K65M) Cosmos 2150
2604	1991-06-13 15:41:00+00:00	Tsyklon-3 Cosmos 2151
2605	1991-06-25 13:20:00+00:00	Cosmos-3M (11K65M) Ta??foun n†59

	Rocket_Status	Price	Mission_Status	Country	Year
0	StatusRetired	NaN	Success	RUS	1957
1	StatusRetired	NaN	Success	RUS	1957
2	StatusRetired	NaN	Failure	USA	1957
3	StatusRetired	NaN	Success	USA	1958
4	StatusRetired	NaN	Success	USA	1958
...	...	...	...	...	...
2601	StatusRetired	NaN	Success	RUS	1991
2602	StatusRetired	450.00	Success	USA	1991
2603	StatusRetired	NaN	Success	RUS	1991
2604	StatusRetired	NaN	Success	RUS	1991
2605	StatusRetired	NaN	Failure	RUS	1991

[2432 rows x 9 columns]

```
[137]: px.pie(
       title = 'Total Number of Launches (USSR vs USA)',
       values = ussr_vs_usa['Country'].value_counts(),
```



```
names = ussr_vs_usa['Country'].value_counts().index,
)
```

Total Number of Launches (USSR vs USA)



17.2 Create a Chart that Shows the Total Number of Launches Year-On-Year by the Two Superpowers

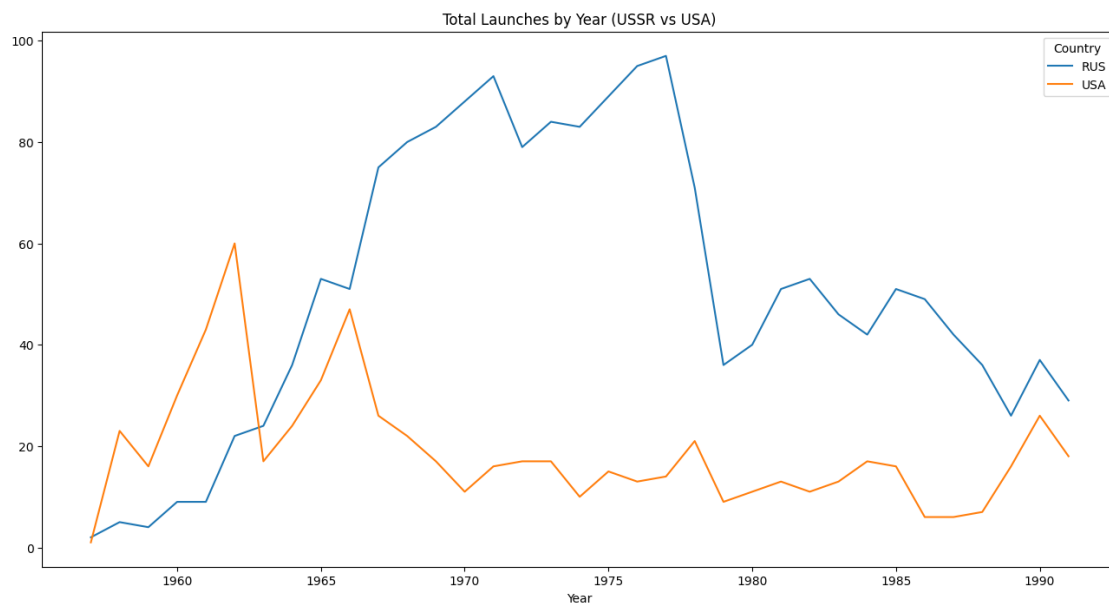
```
[138]: cold_war_launch_by_year = ussr_vs_usa.
        ↳groupby(['Year', 'Country'])['Mission_Status'].count().unstack(level=1)
cold_war_launch_by_year
```

```
[138]: Country  RUS  USA
Year
1957         2    1
1958         5   23
1959         4   16
1960         9   30
1961         9   43
1962        22   60
1963        24   17
1964        36   24
1965        53   33
1966        51   47
1967        75   26
1968        80   22
1969        83   17
1970        88   11
1971        93   16
1972        79   17
1973        84   17
1974        83   10
1975        89   15
1976        95   13
1977        97   14
```

1978	71	21
1979	36	9
1980	40	11
1981	51	13
1982	53	11
1983	46	13
1984	42	17
1985	51	16
1986	49	6
1987	42	6
1988	36	7
1989	26	16
1990	37	26
1991	29	18

```
[139]: cold_war_launch_by_year.plot(
        figsize = (16,8),
        title = 'Total Launches by Year (USSR vs USA)'
    )
```

```
[139]: <Axes: title={'center': 'Total Launches by Year (USSR vs USA)'}, xlabel='Year'>
```



17.3 Chart the Total Number of Mission Failures Year on Year.

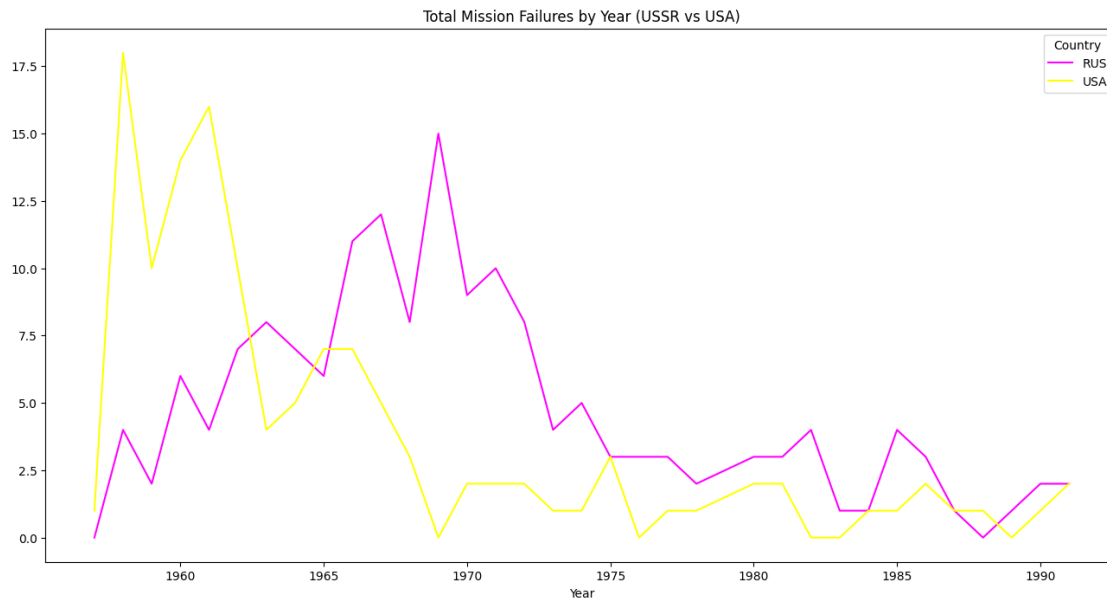
```
[140]: cold_war_mission_failures_by_year = ussr_vs_usa[ussr_vs_usa['Mission_Status']!=
        ↳='Success'].groupby(['Year', 'Country'])['Mission_Status'].count().
        ↳unstack(level=1)
cold_war_mission_failures_by_year.fillna(0, inplace=True)
cold_war_mission_failures_by_year
```

```
[140]: Country    RUS    USA
Year
1957         0.00    1.00
1958         4.00   18.00
1959         2.00   10.00
1960         6.00   14.00
1961         4.00   16.00
1962         7.00   10.00
1963         8.00    4.00
1964         7.00    5.00
1965         6.00    7.00
1966        11.00    7.00
1967        12.00    5.00
1968         8.00    3.00
1969        15.00    0.00
1970         9.00    2.00
1971        10.00    2.00
1972         8.00    2.00
1973         4.00    1.00
1974         5.00    1.00
1975         3.00    3.00
1976         3.00    0.00
1977         3.00    1.00
1978         2.00    1.00
1980         3.00    2.00
1981         3.00    2.00
1982         4.00    0.00
1983         1.00    0.00
1984         1.00    1.00
1985         4.00    1.00
1986         3.00    2.00
1987         1.00    1.00
1988         0.00    1.00
1989         1.00    0.00
1990         2.00    1.00
1991         2.00    2.00
```

```
[141]: cold_war_mission_failures_by_year.plot(
        figsize = (16,8),
        title = 'Total Mission Failures by Year (USSR vs USA)',
```

```
colormap='spring'
)
```

```
[141]: <Axes: title={'center': 'Total Mission Failures by Year (USSR vs USA)'},
      xlabel='Year'>
```



17.4 Chart the Percentage of Failures over Time

Did failures go up or down over time? Did the countries get better at minimising risk and improving their chances of success over time?

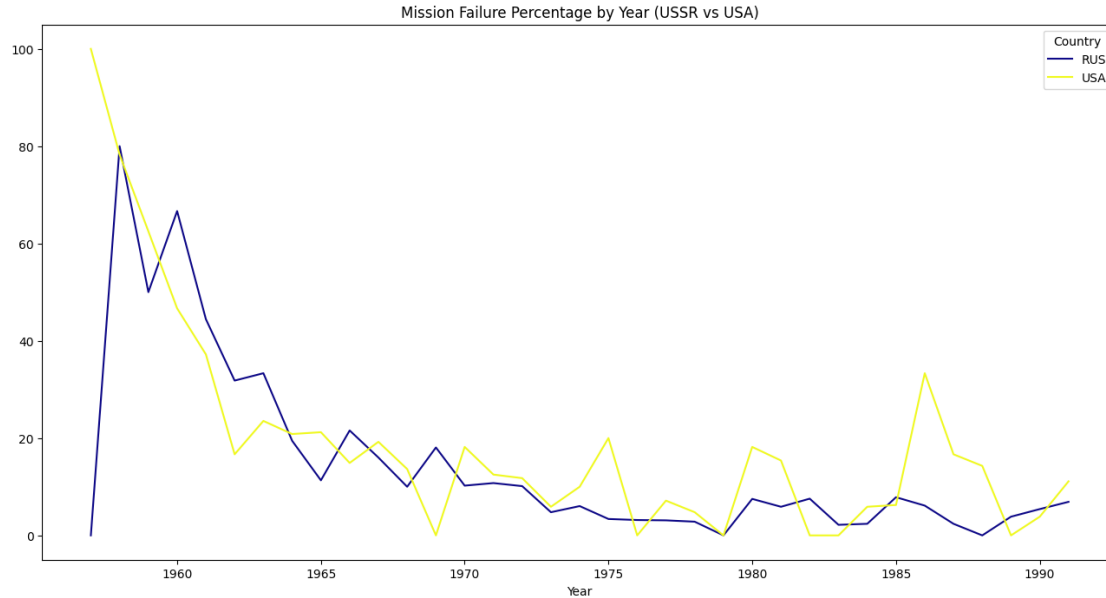
```
[142]: cold_war_failpct_by_year = cold_war_mission_failures_by_year /
      ↪ cold_war_launch_by_year * 100
cold_war_failpct_by_year.fillna(0, inplace=True)
cold_war_failpct_by_year
```

```
[142]: Country  RUS    USA
Year
1957      0.00  100.00
1958     80.00   78.26
1959     50.00   62.50
1960     66.67   46.67
1961     44.44   37.21
1962     31.82   16.67
1963     33.33   23.53
1964     19.44   20.83
1965     11.32   21.21
```

1966	21.57	14.89
1967	16.00	19.23
1968	10.00	13.64
1969	18.07	0.00
1970	10.23	18.18
1971	10.75	12.50
1972	10.13	11.76
1973	4.76	5.88
1974	6.02	10.00
1975	3.37	20.00
1976	3.16	0.00
1977	3.09	7.14
1978	2.82	4.76
1979	0.00	0.00
1980	7.50	18.18
1981	5.88	15.38
1982	7.55	0.00
1983	2.17	0.00
1984	2.38	5.88
1985	7.84	6.25
1986	6.12	33.33
1987	2.38	16.67
1988	0.00	14.29
1989	3.85	0.00
1990	5.41	3.85
1991	6.90	11.11

```
[143]: cold_war_failpct_by_year.plot(
        figsize = (16,8),
        title = 'Mission Failure Percentage by Year (USSR vs USA)',
        colormap='plasma'
    )
```

```
[143]: <Axes: title={'center': 'Mission Failure Percentage by Year (USSR vs USA)'},
        xlabel='Year'>
```



18 For Every Year Show which Country was in the Lead in terms of Total Number of Launches up to and including including 2020)

Do the results change if we only look at the number of successful launches?

```
[144]: launches_by_country_per_year = df_data.groupby(['Year', 'Country'])['Mission_Status'].count().unstack(level=1)
launches_by_country_per_year.fillna(0, inplace=True)
launches_by_country_per_year
```

```
[144]: Country AUS BRA CHN FRA IND IRN ISR JPN KAZ KEN KIR KOR NZL \
Year
1957 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 2.00 0.00 0.00 0.00 0.00
1958 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 5.00 0.00 0.00 0.00 0.00
1959 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 4.00 0.00 0.00 0.00 0.00
1960 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 9.00 0.00 0.00 0.00 0.00
1961 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 7.00 0.00 0.00 0.00 0.00
...
2016 0.00 0.00 22.00 11.00 7.00 0.00 1.00 4.00 11.00 0.00 0.00 0.00 0.00
2017 0.00 0.00 18.00 11.00 5.00 1.00 0.00 7.00 13.00 0.00 0.00 0.00 1.00
2018 0.00 0.00 39.00 11.00 7.00 0.00 0.00 6.00 9.00 0.00 0.00 0.00 3.00
2019 0.00 0.00 34.00 9.00 6.00 3.00 0.00 2.00 13.00 0.00 0.00 0.00 6.00
2020 0.00 0.00 22.00 2.00 0.00 2.00 1.00 3.00 6.00 0.00 0.00 0.00 3.00

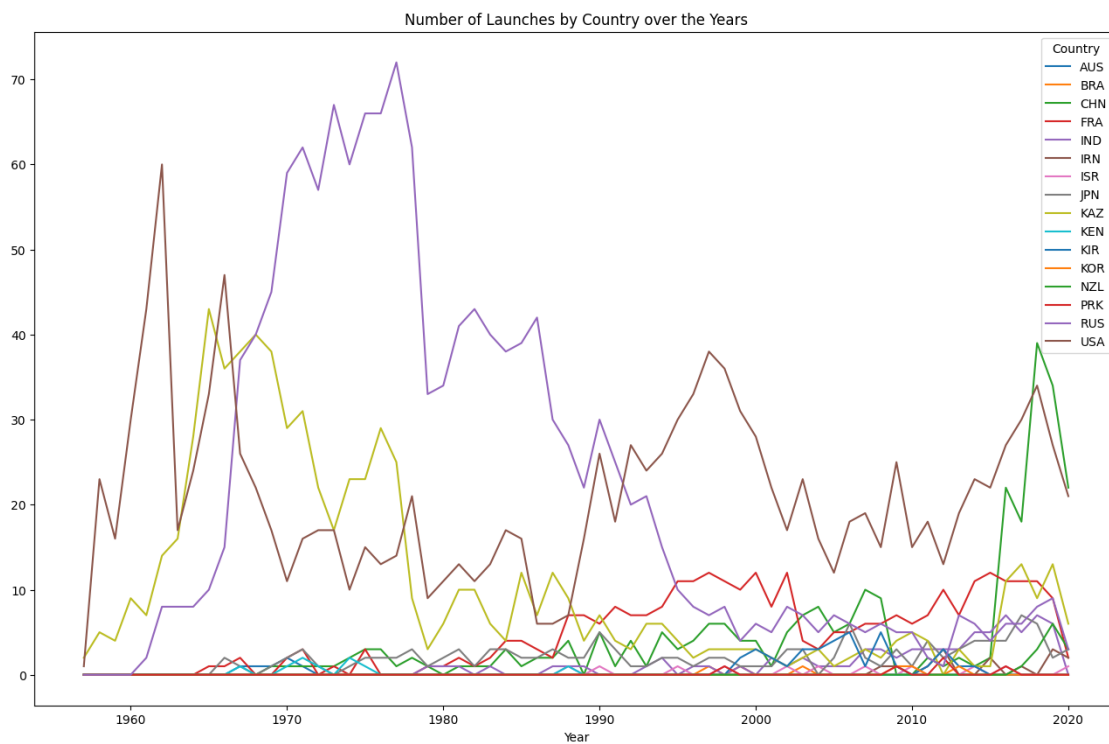
Country PRK RUS USA
```

Year			
1957	0.00	0.00	1.00
1958	0.00	0.00	23.00
1959	0.00	0.00	16.00
1960	0.00	0.00	30.00
1961	0.00	2.00	43.00
...	...	...	...
2016	1.00	6.00	27.00
2017	0.00	6.00	30.00
2018	0.00	8.00	34.00
2019	0.00	9.00	27.00
2020	0.00	3.00	21.00

[64 rows x 16 columns]

```
[145]: launches_by_country_per_year.plot(
        title = 'Number of Launches by Country over the Years',
        figsize = (16,10),
    )
```

```
[145]: <Axes: title={'center': 'Number of Launches by Country over the Years'},
        xlabel='Year'>
```



19 Create a Year-on-Year Chart Showing the Organisation Doing the Most Number of Launches

Which organisation was dominant in the 1970s and 1980s? Which organisation was dominant in 2018, 2019 and 2020?

```
[146]: launches_by_organisation_per_year = df_data.groupby(['Year',
↳ 'Organisation'])['Mission_Status'].count().unstack(level=1)
launches_by_organisation_per_year.fillna(0, inplace=True)
launches_by_organisation_per_year
```

```
[146]: Organisation  AEB  AMBA  ASI  Arianespace  Arm??e de l'Air  Blue Origin  \
Year
1957              0.00  0.00  0.00              0.00              0.00              0.00
1958              0.00  7.00  0.00              0.00              0.00              0.00
1959              0.00  0.00  0.00              0.00              0.00              0.00
1960              0.00  0.00  0.00              0.00              0.00              0.00
1961              0.00  0.00  0.00              0.00              0.00              0.00
...
2016              0.00  0.00  0.00              11.00              0.00              4.00
2017              0.00  0.00  0.00              11.00              0.00              1.00
2018              0.00  0.00  0.00              11.00              0.00              2.00
2019              0.00  0.00  0.00              9.00              0.00              3.00
2020              0.00  0.00  0.00              4.00              0.00              0.00
```

```
Organisation  Boeing  CASC  CASIC  CECLES  ...  SpaceX  Starsem  ULA  \
Year
1957              0.00  0.00  0.00  0.00  ...  0.00  0.00  0.00
1958              0.00  0.00  0.00  0.00  ...  0.00  0.00  0.00
1959              0.00  0.00  0.00  0.00  ...  0.00  0.00  0.00
1960              0.00  0.00  0.00  0.00  ...  0.00  0.00  0.00
1961              0.00  0.00  0.00  0.00  ...  0.00  0.00  0.00
...
2016              0.00  22.00  0.00  0.00  ...  9.00  0.00  12.00
2017              0.00  16.00  1.00  0.00  ...  18.00  0.00  8.00
2018              0.00  37.00  0.00  0.00  ...  21.00  0.00  8.00
2019              0.00  27.00  0.00  0.00  ...  13.00  0.00  5.00
2020              0.00  19.00  0.00  0.00  ...  14.00  0.00  4.00
```

```
Organisation  US Air Force  US Navy  UT  VKS RF  Virgin Orbit  Yuzhmash  \
Year
1957              0.00      1.00  0.00      0.00              0.00      0.00
1958              2.00     12.00  0.00      0.00              0.00      0.00
1959             10.00      4.00  0.00      0.00              0.00      0.00
1960             21.00      0.00  0.00      0.00              0.00      0.00
1961             23.00      0.00  0.00      0.00              0.00      0.00
...
...              ...      ...  ...      ...              ...      ...
```


2016	0.00	0.00	0.00	4.00	0.00	0.00
2017	0.00	0.00	0.00	6.00	0.00	0.00
2018	0.00	0.00	0.00	5.00	0.00	0.00
2019	0.00	0.00	0.00	8.00	0.00	0.00
2020	0.00	0.00	0.00	3.00	1.00	0.00

Organisation	i-Space
Year	
1957	0.00
1958	0.00
1959	0.00
1960	0.00
1961	0.00
...	...
2016	0.00
2017	0.00
2018	0.00
2019	1.00
2020	0.00

[64 rows x 56 columns]

```
[147]: launches_by_organisation_per_year.plot(
        title = 'Number of Launches by Organisation over the Years',
        figsize = (16,10),
    )

plt.ylabel('Number of Missions')
plt.xlim(1957, 2020)
plt.legend(loc="upper right", ncol=4)
plt.show()
```

